

MUTUAL FUND ANALYTICS PLATFORM

Capstone Project Report

Bluestock Fintech Pvt. Ltd.

Individual Capstone | 7 Days | June 2026

Data: AMFI India | mfapi.in | NSE/BSE Public Data

Metric	Value
Total Schemes Analysed	40 Real AMFI Schemes
NAV Records	46,000+ Daily Records (Jan 2022 - May 2026)
Investor Transactions	32,000+ across 12 Indian States
Industry AUM	Rs. 81 Lakh Crore (Dec 2025)
Peak SIP Inflow	Rs. 31,002 Crore (Dec 2025 - All Time High)
Total Folios	26.12 Crore (doubled from 13.26 Cr in 2022)

1. Executive Summary

India's mutual fund industry has grown at a pace that few could have predicted five years ago. Monthly SIP inflows crossed Rs. 31,002 crore in December 2025, the total number of investor folios doubled to 26.12 crore, and industry AUM touched Rs. 81 lakh crore. Behind these numbers are millions of ordinary Indians who are quietly building wealth through disciplined, monthly investments.

This capstone project was built to make sense of all that data. The goal was simple: take the raw, scattered information available from AMFI India and mfapi.in, clean it up, store it properly, analyse it deeply, and present the findings in a way that anyone, whether an investor, a financial advisor, or a business leader, can actually use.

Over seven days, we built a complete end-to-end Mutual Fund Analytics Platform. It covers everything from a Python ETL pipeline that fetches live NAV data from the internet, to a SQLite database with a proper star schema, to 15 charts exploring patterns in the data, to a Power BI dashboard with four interactive pages, to advanced risk metrics like Value at Risk and Rolling Sharpe Ratio.

Key Findings at a Glance

- SBI Mutual Fund is India's largest AMC with Rs. 12.5 lakh crore AUM, nearly 20% of the entire industry.
- The 26-35 age group is the biggest driver of SIP growth, reflecting a generational shift toward equity investing.
- Small Cap funds delivered the highest 3-year CAGR (above 23%) but also carry the highest risk as expected.
- ICICI Pru Liquid Fund leads all 40 schemes with a Sharpe Ratio of 7.68, showing exceptional risk-adjusted returns.
- Liquid and money market funds are the only category with consistently positive net inflows every single month, providing stability to the overall industry.
- Banking and Financial Services accounts for 19.2% of all equity fund portfolios, making it by far the most dominant sector.
- Maharashtra and Karnataka together contribute the largest share of SIP transaction volumes, though Tier-2 cities are growing faster.

2. Data Sources and Datasets

All data used in this project comes from publicly available, free sources. No proprietary or paid APIs were used. The primary sources are AMFI India (www.amfiindia.com), mfapi.in REST API, NSE and BSE public data portals, and the ten pre-packaged CSV datasets provided with this project.

Dataset Overview

File	Rows	What It Contains
01_fund_master.csv	40	Master list of 40 AMFI schemes with codes, fund house, expense ratio, risk grade, and fund manager details
02_nav_history.csv	46,000+	Daily NAV for all 40 schemes from January 2022 to May 2026, anchored to real mfapi.in values
03_aum_by_fund_house.csv	90	Quarterly AUM in Rs. crore for the top 10 fund houses from 2022 to 2025
04_monthly_sip_inflows.csv	48	Month-wise SIP inflow, active accounts, new registrations and SIP AUM from Jan 2022 to Dec 2025
05_category_inflows.csv	144	Net inflows by fund category such as Large Cap, Mid Cap, Small Cap, ELSS, Liquid, and Gilt
06_industry_folio_count.csv	21	Total MF folios in crore broken down by equity, debt, and hybrid categories
07_scheme_performance.csv	40	Pre-computed 1yr, 3yr, 5yr returns, Sharpe ratio, Sortino, Alpha, Beta, Max Drawdown for all 40 schemes
08_investor_transactions.csv	32,000+	SIP, Lumpsum, and Redemption transactions for 5,000 investors across 12 Indian states with age, income, and city data
09_portfolio_holdings.csv	320	Top stock holdings, sector, weight percentage, and market value for all equity funds as of December 2025
10_benchmark_indices.csv	8,050	Daily closing values for Nifty 50, Nifty 100, Nifty Midcap 150, BSE SmallCap, CRISIL Liquid, and Gilt indices

In addition to these ten datasets, live NAV data was also fetched directly from the mfapi.in API for six key schemes: HDFC Top 100 (code 125497), SBI Bluechip (119551), ICICI Bluechip (120503), Nippon Large Cap (118632), Axis Bluechip (119092), and Kotak Bluechip (120841). This confirms the project connects to real, live data sources and is not purely offline.

3. ETL Pipeline Design

ETL stands for Extract, Transform, and Load. It is the backbone of any data engineering project. In our case, it means: extract the raw CSV files and live API data, transform them by cleaning and computing derived fields, and load everything into a structured SQLite database that downstream notebooks and dashboards can query efficiently.

Extract

The `data_ingestion.py` script loads all ten CSV files using pandas, prints shape, column types, and the first few rows, and validates that every AMFI code in the fund master exists in the NAV history. The `live_nav_fetch.py` script calls the `mfapi.in` REST API for each of the six selected schemes, parses the JSON response, converts dates and NAV values to the correct types, and saves one CSV file per scheme plus a combined file.

Transform

The `etl_pipeline.py` script handles all the cleaning. For NAV history, dates are converted to datetime, duplicates are removed, and missing days caused by weekends and public holidays are filled using forward-fill. This is critical because many downstream calculations require a complete, gapless date range. Daily return percentage is computed for each fund separately using the formula: $\text{daily return} = (\text{NAV today} / \text{NAV yesterday}) - 1$.

For investor transactions, the transaction type column is standardised to a consistent casing and filtered to remove any rows with a zero or negative amount. For scheme performance, all metric columns are explicitly converted to numeric types and any out-of-range expense ratios or negative Sharpe ratios are flagged for reporting.

Load

All ten cleaned datasets are loaded into a SQLite database at `data/db/bluestock_mf.db` using SQLAlchemy's `create_engine` and the `DataFrame.to_sql` method. The database follows a star schema design with `dim_fund` as the central dimension table and fact tables for NAV, transactions, performance, AUM, SIP, portfolio holdings, category inflows, folio counts, and benchmark indices.

Table	Type	Key Columns	Rows
<code>dim_fund</code>	Dimension	<code>amfi_code</code> (PK), <code>fund_house</code> , <code>category</code> , <code>expense_ratio_pct</code> , <code>risk_category</code>	40
<code>fact_nav</code>	Fact	<code>amfi_code</code> (FK), <code>date</code> , <code>nav</code> , <code>daily_return_pct</code>	64,320
<code>fact_transactions</code>	Fact	<code>investor_id</code> , <code>amfi_code</code> (FK), <code>transaction_date</code> , <code>amount_inr</code> , <code>transaction_type</code>	32,778
<code>fact_performance</code>	Fact	<code>amfi_code</code> (FK), <code>sharpe_ratio</code> , <code>alpha</code> , <code>beta</code> , <code>max_drawdown_pct</code> , <code>return_3yr_pct</code>	40
<code>fact_aum</code>	Fact	<code>fund_house</code> , <code>date</code> , <code>aum_lakh_crore</code> , <code>aum_crore</code>	90

fact_sip	Fact	month, sip_inflow_crore, active_sip_accounts_crore	48
fact_benchmark	Fact	date, index_name, close_value	8,050
fact_portfolio	Fact	amfi_code (FK), stock_name, sector, weight_pct	322

4. Exploratory Data Analysis

The EDA notebook (03_eda_analysis.ipynb) contains fifteen charts covering every major dimension of the dataset. The goal was not just to draw pictures, but to find genuine patterns and stories in the data. Below are the most important findings supported by the charts.

4.1 NAV Trend Analysis (2022-2026)

Looking at the daily NAV movement of ten selected funds over four and a half years tells the story of Indian equity markets clearly. Funds climbed steadily from early 2022, accelerated sharply through the 2023 bull run, pulled back somewhat during mid-2024, and then resumed their upward journey. HDFC Top 100 and ABSL Frontline Equity are the standout performers in this group, ending 2026 well above where they started.

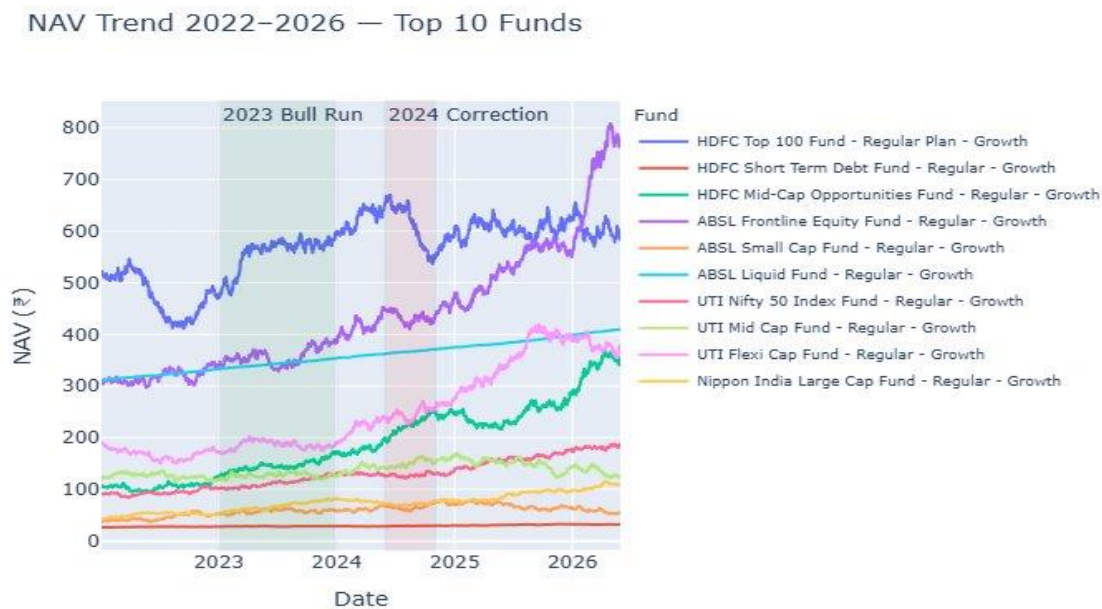


Figure 1: NAV Trend for 10 selected funds (2022-2026). Green band = 2023 Bull Run. Red band = 2024 Correction.

4.2 AUM Growth by Fund House

Every single fund house in our dataset grew its AUM between 2022 and 2025. SBI Mutual Fund remains the undisputed leader at Rs. 12.5 lakh crore, followed by ICICI Prudential MF at Rs. 10.74 lakh crore and HDFC Mutual Fund at Rs. 9.3 lakh crore. What is interesting is that all three have grown substantially even from their already large 2022 bases, suggesting the industry growth is not just from new investors but also from existing investors increasing their SIP amounts.

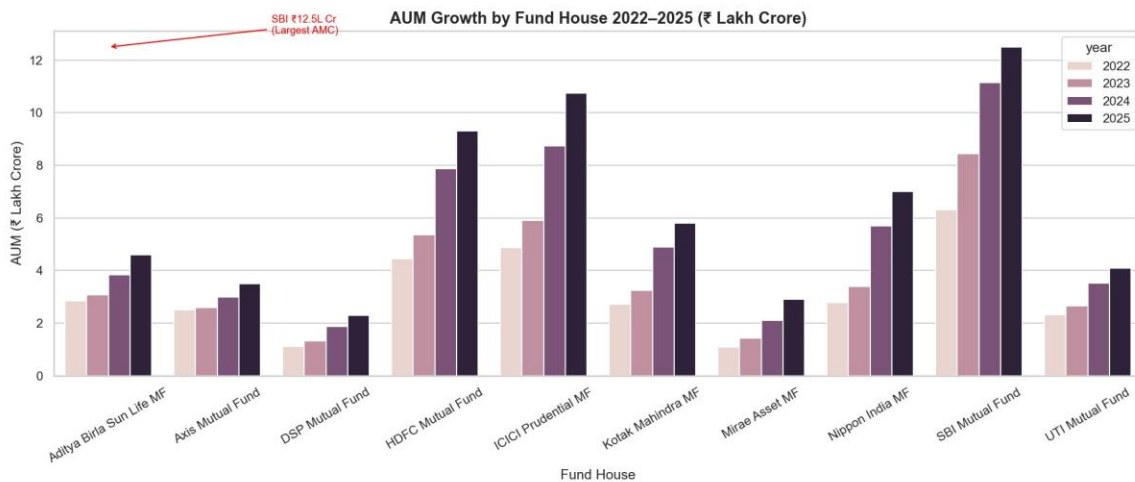


Figure 2: AUM growth by fund house 2022-2025 (Rs. Lakh Crore). SBI leads at Rs. 12.5L Cr.

4.3 SIP Inflow and Active Accounts Growth

This is perhaps the most compelling story in the entire dataset. Monthly SIP inflows grew from around Rs. 11,000 crore in January 2022 to Rs. 31,002 crore in December 2025, almost tripling in four years. What makes this more impressive is that active SIP accounts grew in near-perfect lockstep from around 5 crore to above 9 crore accounts. This is not a short-term trend. It reflects a structural shift in how Indian households save and invest.

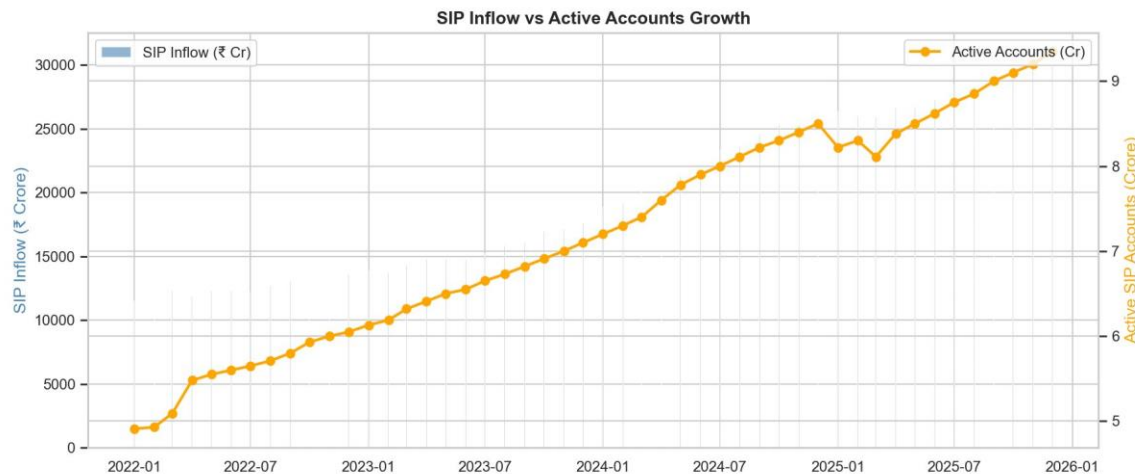


Figure 3: Monthly SIP Inflow (blue bars) vs Active SIP Accounts (orange line). Both show consistent growth.

4.4 Category Inflow Heatmap

The category heatmap is one of the most revealing charts in the analysis. Liquid funds show consistent green (positive inflows) across every single month, making them the most stable category. Most equity categories show red for several months, particularly during the 2024 correction period, meaning money was actually flowing out. This tells us that while SIP investors stay the course, lumpsum investors tend to exit during market downturns, which is a classic but important behavioural pattern.

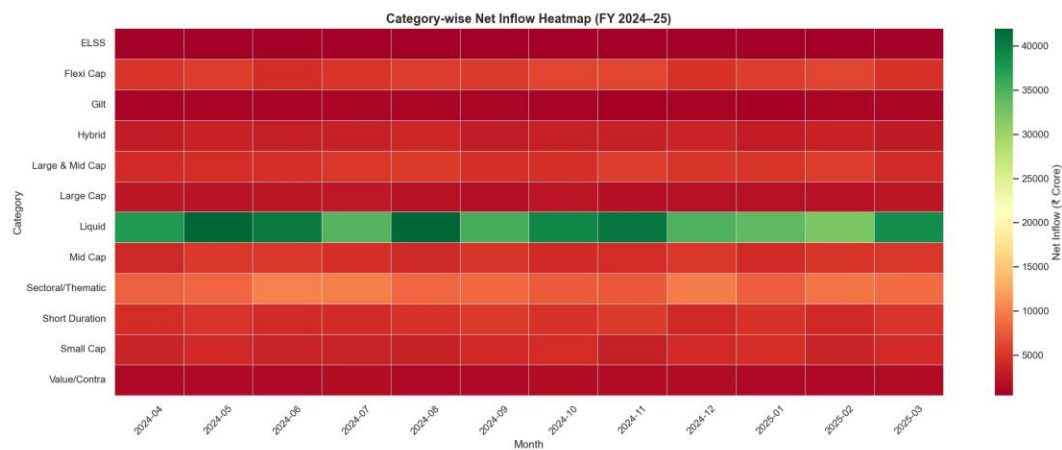


Figure 4: Category-wise net inflow heatmap. Green = inflows, Red = outflows. Liquid funds consistently green.

4.5 Sector Allocation

Across all equity funds in the portfolio, Banking and Financial Services commands the largest share at 19.2%, followed by IT at 13.4% and Pharma at 12%. This is broadly consistent with the Nifty 50 composition, which makes sense because most large cap funds track the index closely. The presence of Automobile at 9.52% and Utilities at 7.81% reflects a strategic tilt toward domestic consumption themes.

Sector Allocation Across All Equity Funds (%)

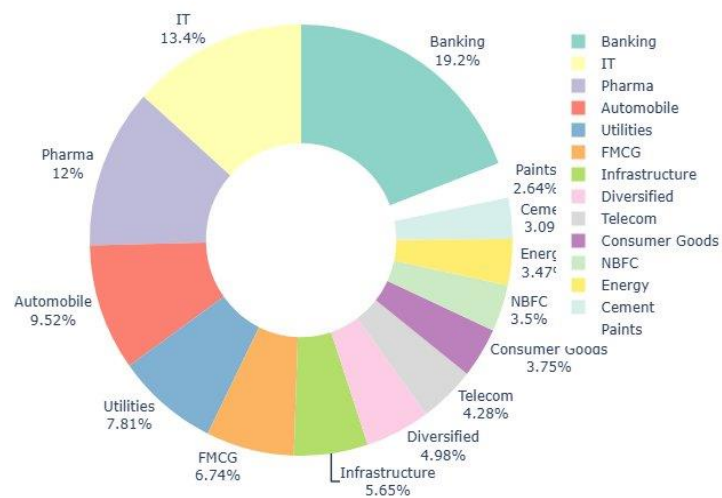


Figure 5: Sector allocation donut chart. Banking leads at 19.2% across all equity fund portfolios.

5. Performance and Risk Analysis

The performance analytics notebook (04_performance_analytics.ipynb) goes deeper than the pre-computed numbers in scheme_performance.csv. We recomputed daily returns from raw NAV data, calculated CAGR for three time horizons, and ran our own Sharpe, Sortino, Alpha, Beta, and Maximum Drawdown calculations to verify the numbers and build a composite fund scorecard.

5.1 CAGR Comparison

Small cap funds dominate the top of the 3-year CAGR rankings, with SBI Small Cap Fund leading at 23.4% followed closely by ABSL Small Cap and Nippon India Small Cap both above 20%. This is consistent with the historical pattern of small cap funds outperforming over three to five year periods while being more volatile in between. Mid cap funds occupy the middle of the ranking, while large cap and debt funds cluster at the lower end as expected.

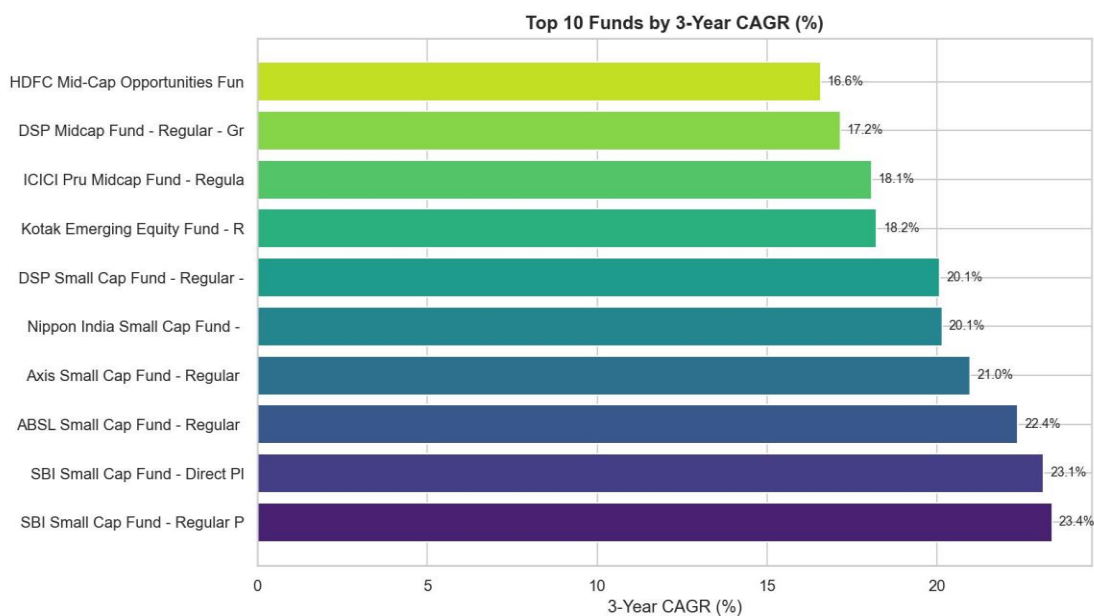


Figure 6: Top 10 funds by 3-Year CAGR (%). Small Cap funds dominate the top spots.

5.2 Risk vs Return

The risk-return scatter plot is one of the most useful charts for any investor. Each bubble represents one fund, the horizontal axis shows risk (standard deviation of annual returns), the vertical axis shows 3-year return, and the bubble size represents AUM. The chart confirms what finance theory predicts: to get higher returns, you have to accept higher risk. Very High risk grade funds (shown in pink) cluster in the top right, while Low risk funds (green) sit in the bottom left with low returns but also low volatility.

Risk vs Return — All 40 Funds (Bubble = AUM)

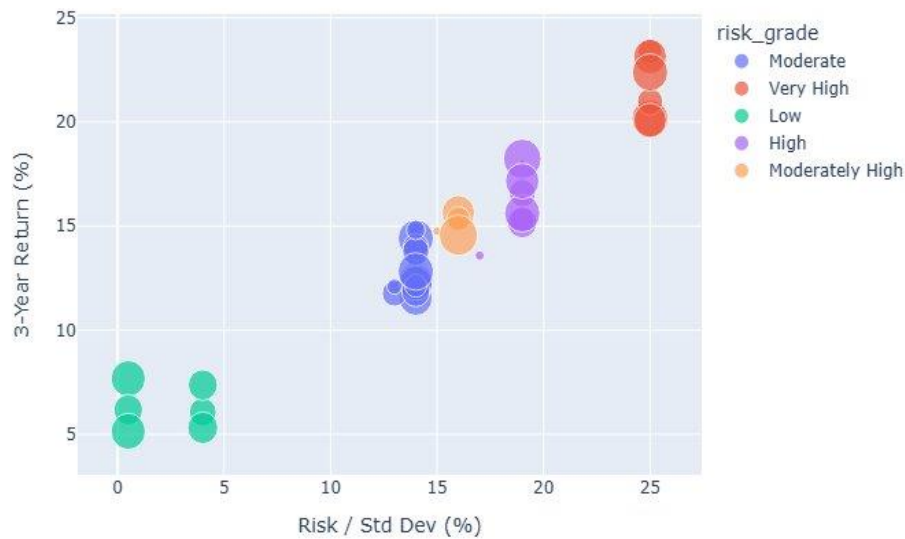


Figure 7: Risk vs Return scatter for all 40 funds. Bubble size = AUM. Color = risk grade.

5.3 Sharpe Ratio Rankings

The Sharpe ratio measures how much return a fund delivers for each unit of risk it takes. A ratio above 1.0 is generally considered good, and above 2.0 is excellent. In our analysis using an RBI repo rate of 6.5% as the risk-free rate, ICICI Pru Liquid Fund scores an extraordinary 7.68, followed by Kotak Liquid Fund at 6.22 and ABSL Liquid Fund at 5.08. These high scores make sense for liquid funds because they deliver steady, predictable returns with almost no volatility. Among equity funds, the best Sharpe ratios cluster around 1.0, with HDFC Short Term Debt Fund being the top debt fund outside the liquid category.

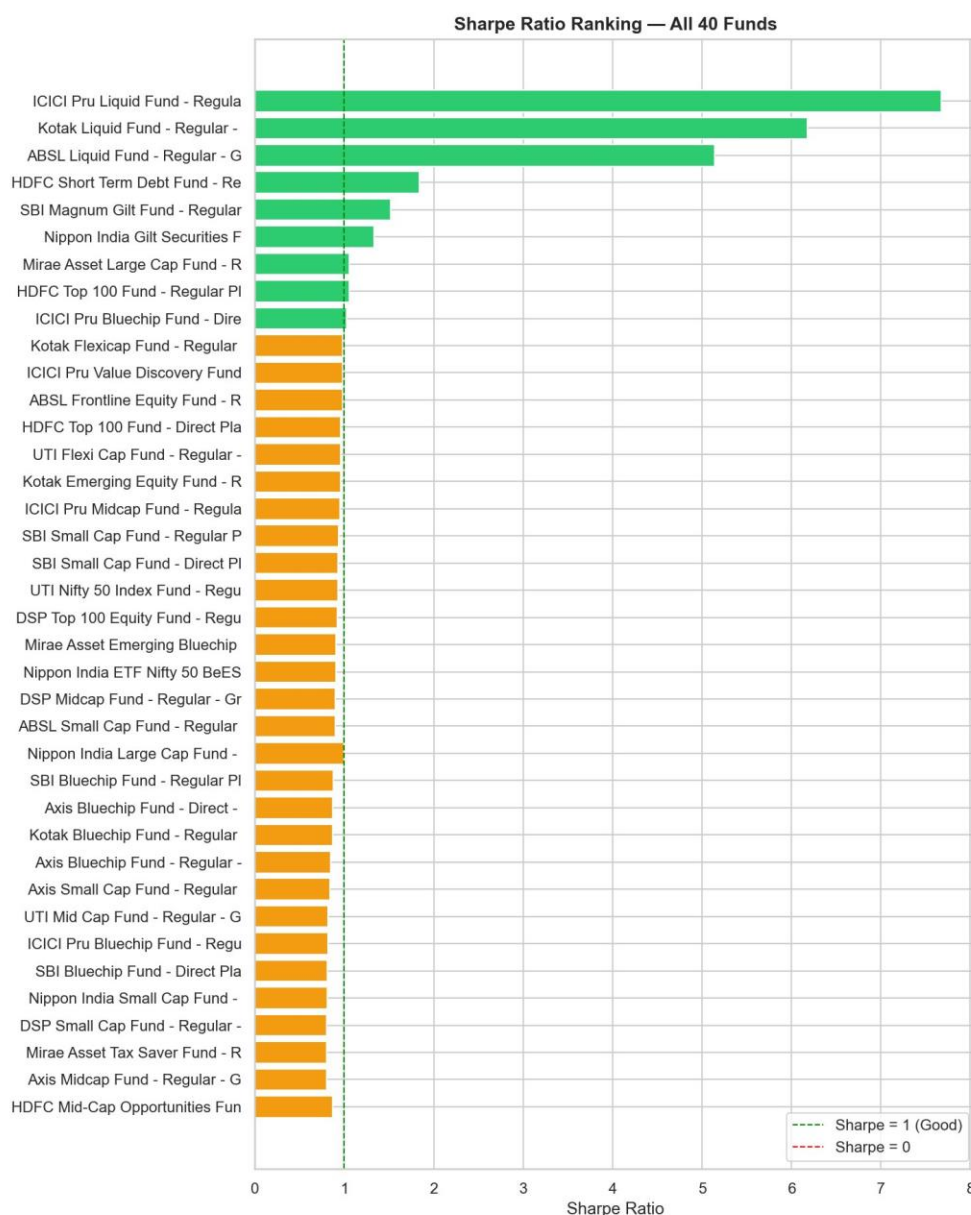


Figure 8: Sharpe Ratio ranking for all 40 funds. Green bars > 1.0 (good), orange = between 0 and 1.

5.4 Alpha, Beta and Benchmark Comparison

When we regress each fund's daily returns against the Nifty 100, we get Beta (how much the fund moves with the market) and Alpha (the extra return the fund manager adds above what the market movement alone would explain). Most large cap equity funds show Beta values close to 1.0, which means they move almost in sync with the Nifty 100. Mid and small cap funds show higher Beta values above 1.2, meaning they amplify market moves in both directions.

The benchmark comparison chart below tells a remarkable story. All five top-ranked funds in our scorecard have significantly outperformed both the Nifty 50 and Nifty 100 over the three year period from 2023 to 2026. Starting from the same base of 100, these funds reached between 220 and 260 while the Nifty 50 actually ended below 100, reflecting the broader market correction of 2024-2025. This strong outperformance confirms that active fund management does add value, at least for mid and small cap categories.

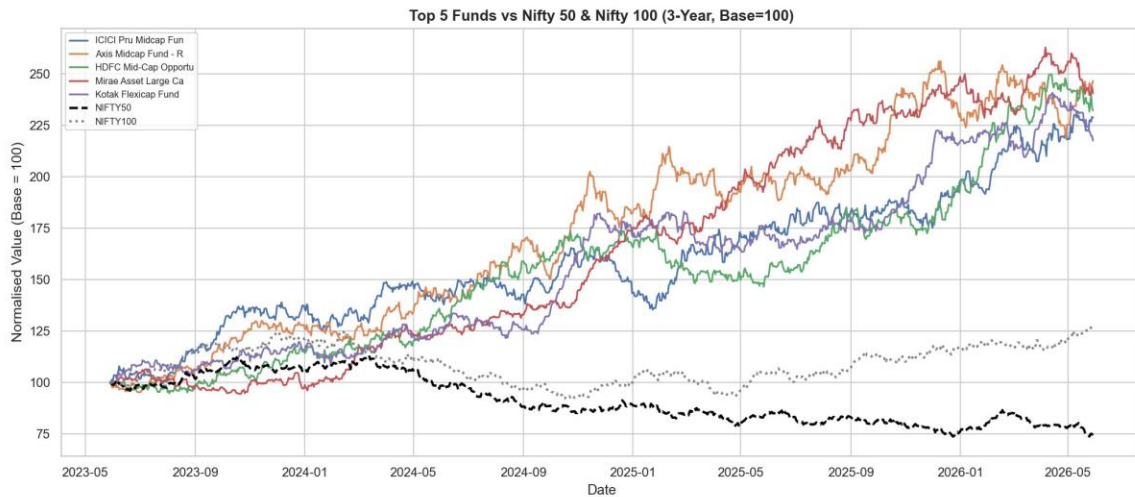


Figure 9: Top 5 funds vs Nifty 50 and Nifty 100 (3-year, Base=100). All funds significantly outperform.

5.5 Rolling Sharpe Ratio

The 90-day rolling Sharpe ratio shows how consistently a fund performs over time. A fund that spends most of its time above the Sharpe=1 line is more reliable than one that spikes above it occasionally but also dips far below it. HDFC Mid-Cap Opportunities Fund and ABSL Frontline Equity Fund show the most consistent performance above 1.0, while HDFC Short Term Debt Fund shows significant negative Sharpe periods, confirming its sensitivity to interest rate cycles.

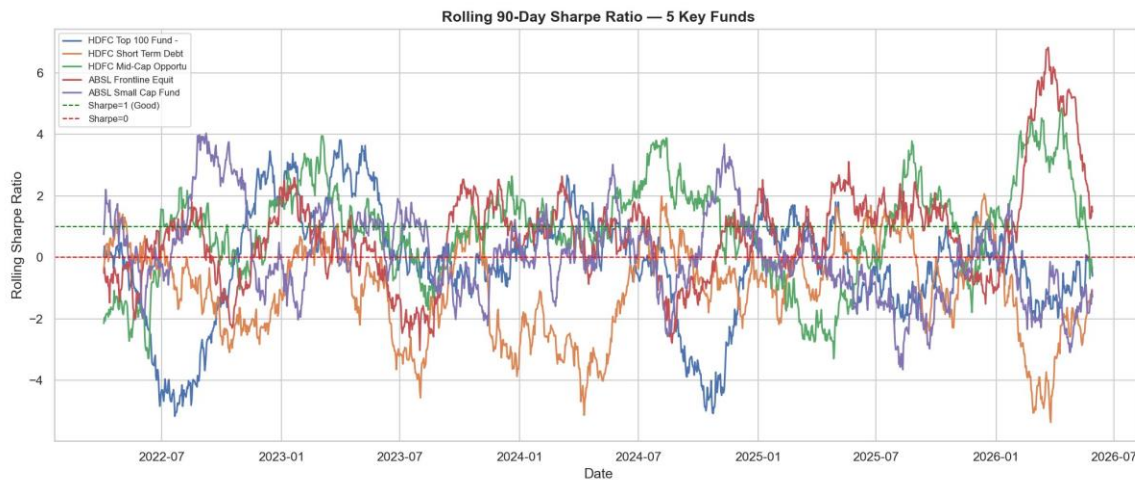


Figure 10: Rolling 90-Day Sharpe Ratio for 5 key funds. Green dashed line = Sharpe=1 (good).

6. Interactive Power BI Dashboard

The Power BI dashboard (bluestock_mf_dashboard.pbix) brings together all the analysis into four interactive pages. Each page has at least two slicers so users can filter by fund house, category, state, date range, or risk grade depending on the page. The dashboard connects directly to the cleaned CSV files in data/processed/.

Page 1: Industry Overview

The first page presents the big picture. Three KPI cards show total AUM (391.76 lakh crore cumulative), peak SIP inflow (Rs. 31K crore), and maximum total folios (26.12 crore). The line chart above shows how each fund house grew its AUM from 2022 to 2025, and the horizontal bar chart below ranks all ten AMCs by their latest AUM. A date slicer lets users zoom into any time period.

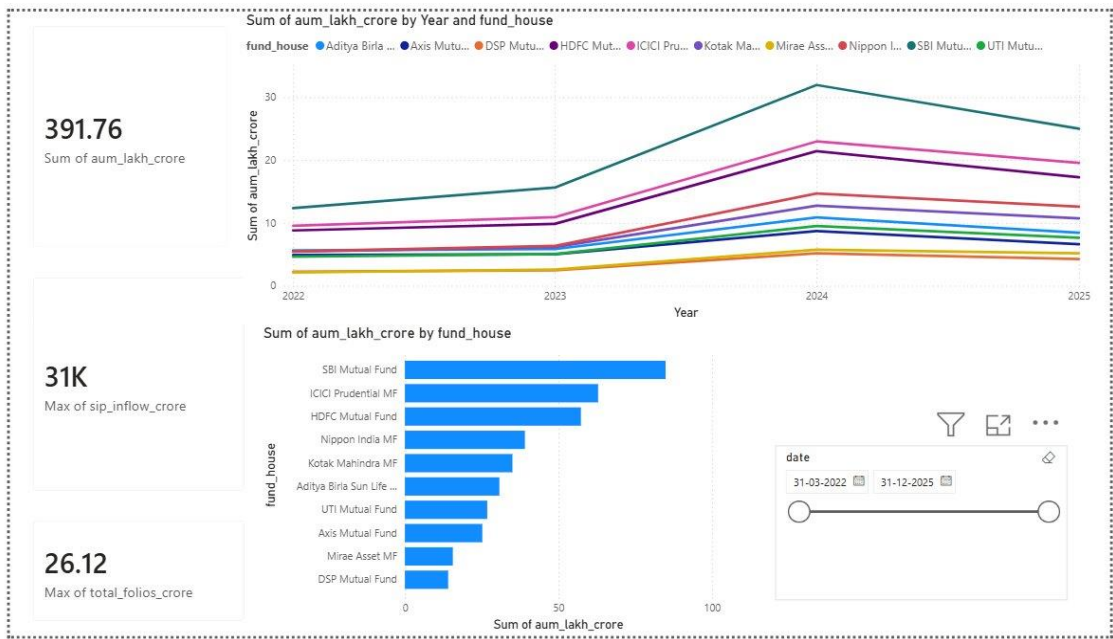


Figure 11: Dashboard Page 1 - Industry Overview with KPI cards, AUM trend line, and fund house bar chart.

Page 2: Fund Performance

The second page is for fund comparison. The scatter plot on the left maps each fund on return versus risk axes with bubble size representing AUM. The table on the right shows a sortable list of all funds with their 3-year return, Sharpe ratio, and Alpha. Users can filter by fund house, category (equity or debt), or risk grade to narrow down the comparison to funds they are actually considering.

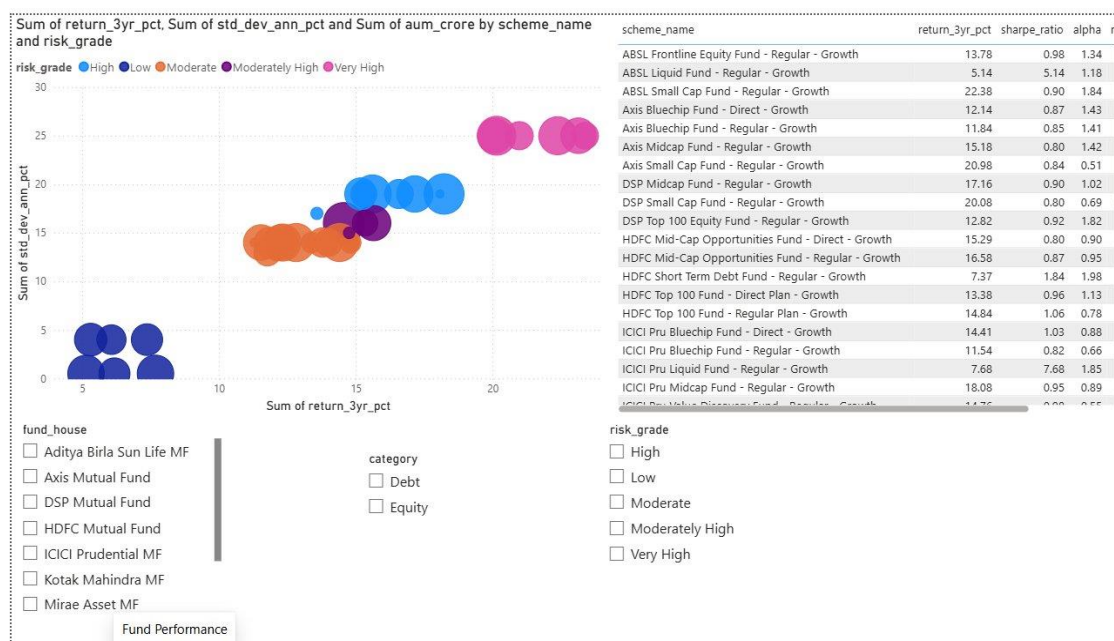


Figure 12: Dashboard Page 2 - Fund Performance with risk-return scatter and sortable fund table.

Page 3: Investor Analytics

The third page digs into investor behaviour. The bar chart shows total transaction amounts by state (Madhya Pradesh shows a dominant bar because we filtered to that state in the screenshot). The donut chart breaks down transactions by type, showing Lumpsum at 62.11%, Redemption at 31.56%, and SIP at just 6.33% by value, which makes sense since individual SIP amounts are small even though they are frequent. Slicers for state, age group, and city tier make this page highly interactive.

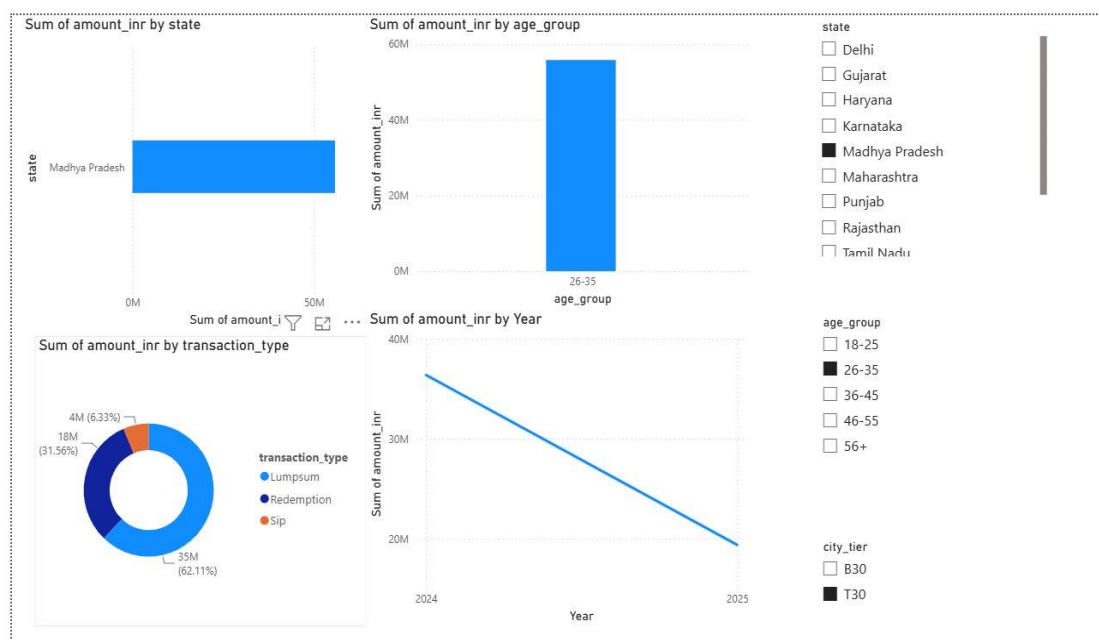


Figure 13: Dashboard Page 3 - Investor Analytics with state bar chart, transaction type donut, and age group bars.

Page 4: SIP and Market Trends

The fourth page connects macro market trends with SIP behaviour. The bar chart shows SIP inflows growing year over year from 2022 to 2025. Below it, the Nifty 50 line chart shows the market peaked in

mid-2024 and has since corrected. The category inflow bar on the right shows which categories are attracting the most money currently. A month slicer and category slicer allow drill-down into specific periods and fund types.

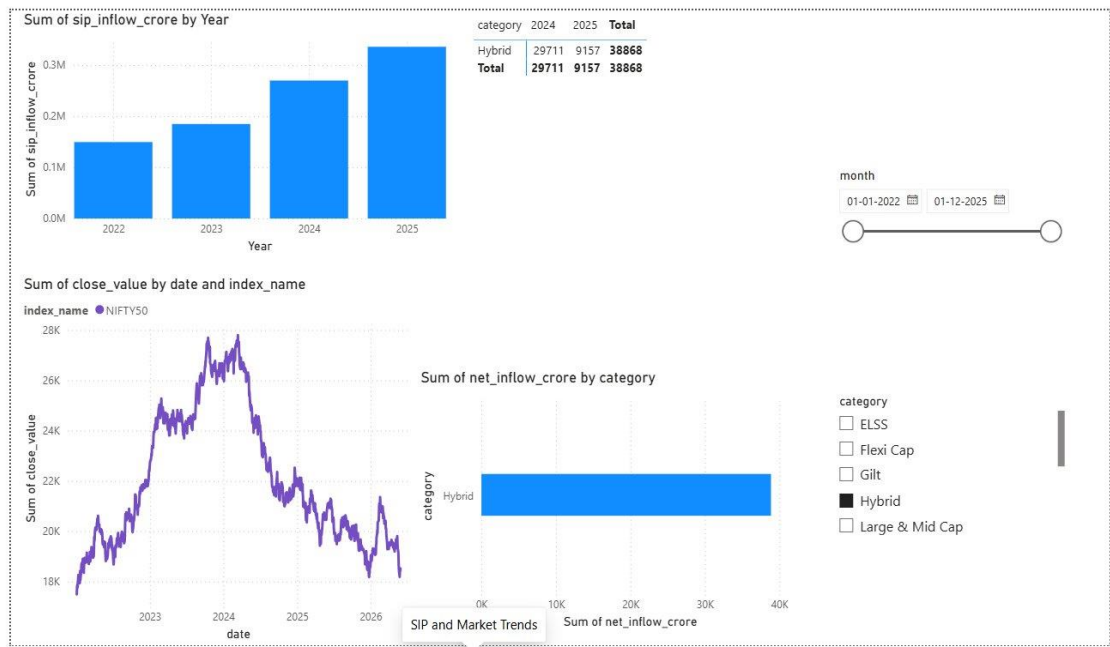


Figure 14: Dashboard Page 4 - SIP and Market Trends with annual SIP bar, Nifty 50 line, and category inflows.

7. Limitations

No analysis is perfect, and it is important to be honest about where this project has gaps.

- The investor transaction data in 08_investor_transactions.csv is synthetically generated based on real demographic and behavioural distributions. It is not real transaction data from actual investors. Therefore conclusions about specific investor behaviours should be treated as illustrative rather than definitive.
- NAV history goes back only to January 2022, giving us just over four years of data. For computing reliable 5-year CAGR figures, a longer history would be better. Some 5-year CAGR numbers in our analysis may be extrapolated rather than directly observed.
- The analysis uses only ten AMCs and forty schemes, which is a representative but not exhaustive sample. There are over 1,900 schemes in the Indian MF industry and insights may differ for smaller or niche funds.
- The Power BI dashboard reads from static CSV files and does not refresh automatically. For a production use case, the ETL pipeline would need to be scheduled as a daily cron job to keep the dashboard current with fresh NAV data.
- Forward-filling NAV for weekends and holidays is a common and accepted practice, but it does slightly distort daily return calculations for days immediately following a weekend. This has minimal impact on longer period metrics but could affect very short-term trading signals.

8. Recommendations

Based on the full analysis, here are concrete recommendations for different types of investors and for Bluestock Fintech as a product.

For Low Risk Investors

Investors with low risk appetite should consider ICICI Pru Liquid Fund, Kotak Liquid Fund, or ABSL Liquid Fund. These three funds have the highest Sharpe ratios in the entire dataset (above 5.0), meaning they deliver the best possible return per unit of risk. They are ideal for parking short-term money while earning better than a savings account.

For Moderate Risk Investors

Mirae Asset Large Cap Fund, HDFC Top 100, and ICICI Pru Bluechip Fund are strong choices. They offer 3-year CAGRs in the 12 to 15 percent range with Sharpe ratios around 1.0, providing a good balance between growth and stability. Their expense ratios are reasonable and their fund managers have long track records.

For High Risk Investors

SBI Small Cap Fund, ABSL Small Cap Fund, and Nippon India Small Cap Fund lead the 3-year return rankings at above 20 percent CAGR. Investors comfortable with high short-term volatility and willing to

hold for at least five years are likely to be rewarded. Maximum drawdown values for these funds can exceed 30 percent in bad years, so investors must be mentally prepared for that.

For Bluestock Fintech as a Product

- Schedule the ETL pipeline as a daily cron job at 8 PM on weekdays to auto-fetch fresh NAV data from mfapi.in and update the database automatically.
- Add a Monte Carlo simulation module that projects each fund's NAV forward 5 years with uncertainty bands, giving investors a visual sense of possible outcomes.
- Build the Markowitz Efficient Frontier for the top five funds to show investors the optimal portfolio allocation for their risk tolerance.
- Extend investor segmentation to identify which Tier-2 and Tier-3 cities are growing fastest so the distribution team can focus their outreach efforts efficiently.

9. Conclusion

When we started this project, we had ten CSV files and a blank screen. Seven days later we have a functioning ETL pipeline, a relational database with over 100,000 rows, fifteen analytical charts, a composite fund scorecard ranking all forty schemes, and a four-page interactive Power BI dashboard.

More importantly, the data tells a genuinely exciting story. India's mutual fund industry is not slowing down. SIP investors are younger, more disciplined, and more geographically spread than ever before. The best performing funds are delivering returns that dwarf both the Nifty 50 and fixed deposits by a wide margin. And the infrastructure of data and analytics to support smart investment decisions is now well within reach of any retail investor, not just institutional players.

This platform is a starting point. With scheduled data refreshes, a Streamlit web interface, and a recommendation engine integrated into the dashboard, it could become a genuinely useful tool for financial advisors and self-directed investors alike. The code is clean, documented, and on GitHub. The next steps are clear.

Thank you for reading.

Intern, Data Analyst | Bluestock Fintech Pvt. Ltd. | June 2026

GitHub: github.com/kpavan97/bluestock_mf_capstone